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MILITARY SPECIFICATION

KETTLES, HEATING, BITUMEN

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of trailer mounted liquified petroleum gas (lpg) and kerosene fueled, bitumen heating kettles. The kettles shall be used for heating, transporting, and pressure discharging bitumen materials used in paving and roofing applications.

1.2 Classification. The kettles shall be of the following types, styles, and sizes as specified (see 6.2).

Type I - Bottom Fired (Road Kettle).
Type II - Tube Fired (Roofing Kettle).

Style A - Lpg as burner and engine fuel.
Style B - Kerosene (light oil) as burner fuel and gasoline as engine fuel.
Style C - Lpg as burner fuel and gasoline as engine fuel.

Size 80 - Melting tank capacity of 80 gallons.
Size 165 - Melting tank capacity of 165 gallons.
Size 250 - Melting tank capacity of 250 gallons.
Size 350 - Melting tank capacity of 350 gallons.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

MILITARY

MIL-D-771 - Distributor, Bitumen, and Water, Packaging of.

STANDARDS

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF INTERIOR (BUREAU OF MINES)

IC-8333 - Ringelmann Smoke Chart.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

(Application for copies of IC-8333, OSHA, and DoT should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

CODE OF FEDERAL REGULATIONS (CFR)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Title 40, Protection of Environment

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(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J534 - Lubrication Fittings.
J551 - Standard for Measurement of Electromagnetic Radiation from Motor Vehicles (20-1,000 MHz).
J560 - Seven-Conductor Electrical Connector for Truck Trailer Jumper Cable.
J588 - Turn Signal Lamps.
J1067 - Standard for Seven-Conductor Jacketed Cable for Truck and Trailer Connections.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 644 - Container Assemblies for LP-Gas.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Kettles, hereinafter referred to as kettles, shall consist essentially of a melting tank with an outside furnace shell, removable torch-type fuel burner(s), fuel tank(s), and all necessary controls for normal

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operation. Type I kettles shall be bottom fired and equipped with a pressurized hand-spray attachment. Type II kettles shall be tube fired and equipped with a submerged pump and delivery system (tubing, connectors, elbows, etc.) for roofing applications. The kettles and all components shall be mounted on a pneumatic-tired trailer furnished with not less than two wheels.

3.1.1 Standard commercial product. The kettles shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the kettles being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturers' catalogs, or brochures, and represents the latest production model.

3.2. First article. When specified (see 6.2), the contractor shall furnish a complete kettle for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Construction. The kettles shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

3.4 Interchangeability. All kettles of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.6 Safety. All parts subject to high operational temperatures, that are of such a nature or are so located as to be or become a hazard to the safety of the operating personnel, shall be insulated, enclosed, or fully guarded to the extent necessary to eliminate the hazard. The kettles shall comply with applicable OSHA regulations in effect at time of manufacture.

3.7 Air pollution control. The kettles shall comply with applicable EPA Regulations governing control of air pollution. Smoke discharge density, in accordance with the Ringelmann Smoke Chart, shall be not greater than 20 percent at operating temperature.

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3.8 Performance.

3.8.1 Type I kettle. Type I kettle shall be capable of heating all types of bituminous cements and tar to a spraying temperature of 350 degrees Fahrenheit (^{o}F) $\pm 25^{o}$ and discharging heated product through the hand spray attachment nozzle at a rate of not less than 10 gallons per minute (gpm) at 30 pound-force per square inch (psi). The pump system, when not a submerged type, shall be so designed as to pump cleaning solvent through the hand spray attachment without the solvent entering the melting tank. Type I kettle shall be capable of heating not less than 50 percent of its volume within one hour.

3.8.2 Type II kettle. Type II kettle shall be capable of heating all types of bituminous roofing cements and tar to a temperature of 550°F $\pm$ 25° and be capable of discharging heated product from a submerged positive displacement rotary gear type pump at a rate of not less than 35 gpm through 1-1/2 inch thinwall tubing to a height of not less than 75 feet. Type II kettle shall be capable of heating not less than 50 percent of its volume within one hour.

3.9 Kettle.

3.9.1 Type I melting tank. Type I melting tank shall be constructed of sheet steel of not less than 0.1345 inch (U.S. revised standard gage No. 10) thickness. All seams shall be welded and leaktight. Splash plates shall be provided on the top edges around the inner rim of the melting tank. An outlet shall be provided at the lowest point in the tank and shall be fitted with a 1-1/2-inch minimum size draw off cock. The melting tank shall be so designed that only melted material can enter the area of the inlet to the pump. A dial type bimetallic thermometer calibrated from 50°F to 350°F shall be provided. The dial shall be protected from inadvertent damage. The thermometer temperature-sensing element shall be located at the point of greatest heat concentration. The liquid capacity of the melting tank shall be at least 4 percent greater than the rated capacity to provide for the expansion of the bitumen.

3.9.2 Type II melting tank. Type II melting tank shall be constructed of sheet steel of not less than 0.1046 inch (U.S. revised standard gage No. 12) thickness. All seams shall be welded and leaktight. Splash plates shall be provided on the top edges around the inner rim of the melting tank. A 2-inch minimum draw off cock shall be provided in the tank. Type II kettle shall be equipped with a tube fired heating system (see 3.9.7.2). The melting tank shall be so designed that only melted material can enter the area of the inlet to the pump or the pump shall be screened. A dial type bimetallic thermometer calibrated from 500°F to not less than 750°F shall be provided. The thermometer sensing element shall be located at the point of greatest heat concentration. The liquid capacity of the melting tank shall be at least 8 percent greater than the rated capacity listed in 1.2 to allow for expansion of the bitumen.

3.9.3 Insulation.

3.9.3.1 Type I insulation. Type I kettle melting compartment shall be thermally insulated from the kettle shell. Thermal insulation shall reduce radiated heat from the kettle shell to a suitable safe working level. Type I kettles shall be insulated with, as a minimum, a refractory type blanket not

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less than 1-inch thick having a density of not less than 6 pounds (lb) per cubic foot. The blanket shall be mechanically secured to the shell.

3.9.3.2 Type II insulation. Type II kettles shall be insulated with as a minimum a 1,000°F 1-inch thick fiberglass blanket. The blanket shall be securely attached. The insulation shall not disintegrate under vibration or settle and leave voids during the life of the kettle.

3.9.4 Kettle shells (outer shells). Shells for type I kettles shall be constructed of not less than 0.0747 inch (U.S. revised standard gage No. 14) nominal thickness and for type II kettles shall be constructed of not less than 0.0478 inch (U.S. revised standard gage No. 18) nominal thickness. All seams shall be welded. The shell top edges shall be flanged to provide a support for the melting tank. The kettle shell sides shall not be vented. Type I kettle shell shall be equipped with a burner compartment for bottom fired type heating system and provided with a vent stack.

3.9.5 Covers.

3.9.5.1 Type I cover. Type I kettle shall have a hinged cover, made of steel not less than 0.0747 inch (U.S. revised standard gage No. 14) in thickness, which shall be provided on the melting tank. A device shall be provided that will hold the cover in the fully open and closed position. The cover shall be provided with handles and with a latch to hold the cover in the closed position. Cover shall have tight fit.

3.9.5.2 Type II cover. Type II kettles shall have the same cover as specified for type I kettles except that flash vents shall be provided on type II kettles. Vents providing a minimum of 100 square inches of area shall be provided for kettles up to 200 gallon capacity and 200 square inches for kettles over 200 gallon rated capacity. Flash vents shall conform to current OSHA regulations.

3.9.6 Grille. When specified (see 6.2), a removable grille which will support a 55-gallon drum, filled with bitumen, while it is being emptied into the melting tank shall be provided under the cover for type I kettles of 165 gallons and larger.

3.9.7 Heating system.

3.9.7.1 Type I heating system. Type I kettles shall be bottom fired. The furnace shell shall be provided with a burner box, baffle plates, and a vent stack. The baffle plates shall be arranged to provide heat circulation completely around the sides and the bottom of the melting tank. The vent stack shall be provided with a hinged cover.

3.9.7.2 Type II heating system. Type II kettles shall be tube fired. Tube fired system shall consist of the fire tube(s), manifold(s), and return tube(s). Length, diameter, and design of tubes shall be the manufacturer's standard design.

3.9.8 Burner(s) system.

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3.9.8.1 Type I burner(s). The burner(s) shall be of the removable torch type. Style A and C shall be designed for use with vapor from lpg. Style B shall be designed to burn kerosene (light fuel oil). The burner(s) shall be equipped with a windshield or baffle, all necessary fittings, controls, and a fuel line. Style B fuel line shall be an oil-and-heat-resistant flexible hose and shall be equipped with suitable joint fittings. Style A and C burner(s) shall be equipped with a pilot control system, designed so the pilot light must be lit before the main fuel valve will open, and an automatic temperature control system (see 3.9.9).

3.9.8.2 Type II burner(s). Type II burner(s) shall be of same design as type I except that type II shall have a burner box located at the entry of the fire tube(s). Windshield or baffle need not be furnished for type II. An automatic temperature control system is required for style A and C (see 3.9.9).

3.9.9 Automatic temperature control system. Kettles equipped with lpg burners (style A and C) shall be equipped with a fully self contained, 100 percent fail-safe automatic temperature control system to maintain tank temperature at an adjustable level in a range of 50°F to 350°F for type I kettles and 50°F to 550°F for type II kettles. The thermostat supplied shall be hermetically sealed and modified, if necessary, by the vendor so that the maximum temperature of the material heated cannot exceed 350°F for type I and 550°F for type II. Pilot light-out protection shall be installed. Battery operated controls shall not be furnished. The capillary tube holder shall be mounted in the tank with sensor extending down into the liquid for accurate temperature measurement. This capillary tube holder shall not go through the fire chamber. All automatic controls shall be completely enclosed by a protective and lockable enclosure with hinged access door.

3.9.10 Lpg cylinder(s) or tank. Lpg fueled kettles shall have an approved, not less than 100 lb capacity, pressure vessel(s) or tank mounted on the trailer. The mounting for removable pressure vessels shall have adequate holddown devices to secure the cylinder(s) under all driveable surface conditions and enable the cylinders to be readily installed or removed. Cylinders shall conform to UL 644. Contractor shall supply all necessary hardware, valves, and fittings to connect gas supply to burner(s).

3.9.11 Kerosene fuel tank. Kerosene fueled kettles shall have a fuel tank with a capacity for not less than four hours of operation. The tank shall be equipped with a 2-inch filler opening having a cap with a built-in, manually operated, air-release valve; an air pressure gage calibrated in 5-lb increments from 0 to 60 lb; hand-operated air pump; and an automatic relief valve set for not more than 50 psi. The air pressure gage shall be marked to indicate the normal operating pressure range. The cylinder of the air pump shall be located inside the fuel tank. The outlet to the fuel burner shall be equipped with a globe valve. The valve shall have threaded connections. The tank shall be mounted on the chassis of the trailer and shall be protected from the heat radiated by the melting tank and the burner(s). A caution plate shall be located near the fuel tank filler opening. The caution plate shall read: FILL WITH DIESEL FUEL OR KEROSENE ONLY.

3.9.12 Pump.

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3.9.12.1 Type I pump. Type I kettles shall be supplied with a removable engine-driven bitumen pump. The pump shall deliver standard commercial grades of paving asphalt having a penetration of 85-100, such as Chevron asphalt grade AC-10 or equivalent, heated to a temperature of 350°F through the spray nozzle at a rate of not less than 10 gpm at not less than 30 psi. A bypass relief valve shall be provided for the pump. The pump shall be installed on the kettle and connected in a manner which permits pumping the liquid asphalt through the hand spray attachment. A submerged rotary gear pumping system may be provided. The pump shall be readily removable for service without disconnecting numerous piping. Nonsubmerged pump systems shall be so designed as to pump cleaning solvent through the hand spray attachment without the solvent entering the melting compartment.

3.9.12.2 Type II pump. Type II kettles shall be supplied with a removable engine-driven, submerged, positive displacement, rotary gear type bitumen pump. The pump shall be capable of pumping not less than 35 gpm through 1-1/2 inch thinwall tubing to a height of 75 feet. The pump shall be readily removable for service without disconnecting numerous piping. The pump shall provide means for recirculating heated product through the tubing system.

3.9.12.3 Pump engine. Pump engines shall be the manufacturer's standard commercial engine. The power and speed rating of the engine shall be such that operation of the kettles, under any of the requirements and tests specified herein, will not require horsepower (hp), in excess of the continuous hp rating of the engine at the applicable governed speed.

3.9.12.4 Style A engine. Style A engine shall be suitably equipped to operate on vapor lpg.

3.9.12.5 Style B and style C engines. Style B and C engines shall be equipped with the manufacturer's standard commercial fuel tank and starting systems. The engine shall produce the required performance on unleaded or low-lead content gasoline.

3.9.12.6 Engine housing. Engines shall be enclosed in a metal housing. The housing shall permit access to the engine.

3.9.13 Hand spray attachment. A hand spray attachment shall be provided for type I kettles. The hand spray attachment shall consist of a suitable flexible discharge hose, a spray bar, and a replaceable nozzle. The hose shall be not less than 15 feet long with a 1-inch inside diameter (id). Each end of the hose and the hose end of the spray bar shall be provided with a quick-disconnect coupling or union made of corrosion-resistant material. The spray bar shall be provided with a quick-acting shutoff valve and two handles that minimize heat transfer. One of the handles shall operate the shutoff valve. The spray bar shall be equipped with a replaceable nozzle. The nozzle shall produce a fan-type spray pattern and shall provide even distribution from a height of not less than 10 inches. Provisions shall be made to carry the hose and spray bar on the kettle.

3.9.14 Tubing and connectors. Thinwall tubing, connectors, elbows, and couplings shall be provided in sizes and lengths as specified for roofing applications (see 6.2).

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3.9.15 Fluid level indicator. Kettles shall be provided with a fluid level indicator, such as a dipstick, that will indicate the level of liquid asphalt or pitch within the kettle without the necessity of opening the cover for direct observation.

3.9.16 Emergency stop device. Type II kettles shall be supplied with an emergency stop device that can be operated from the roof by one person. The device shall shut down the engine by using a rope or other suitable means.

3.10 Trailer. Kettles shall be mounted on a trailer having not less than two wheels. Trailer shall be cart-type trailer chassis that shall withstand towing over paved roads at speeds up to 35 miles per hour (mph) to sustain an average speed of 25 mph. The trailer shall comply with applicable DoT Federal Motor Carrier Safety Regulations.

3.10.1 Tongue, lunette, and safety chains. The trailer tongue shall be equipped with a lunette having an eye of not less than 3-inch id and conforming to standard commercial practice. Two safety chains, conforming to DoT Federal Motor Carrier Safety Regulation, Section 393.70, shall be attached to the tongue. Each chain shall extend between 36 and 42 inches beyond the lunette eye and shall be equipped with a safety hook.

3.10.2 Support jacks. The trailer shall be equipped with adjustable and retractable support jacks of sufficient quantity to support the kettle in operating position, under all conditions of loading. When the support jacks are retracted, the clearance between the jacks and the ground shall be not less than the clearance between the axle and the ground. When specified (see 6.2), a tongue mounted screw jack shall be provided.

3.10.3 Suspension. The kettle shall be attached to the axle by either a torsion-coil spring arrangement, or semielliptic leaf springs that shall withstand the imposed static and dynamic loadings. The axle shall have a rated capacity in excess of that imposed when the kettle is filled to capacity with asphalt cement and fuel.

3.10.4 Fenders. The wheels shall be shielded by fenders to restrict water and other road surface substances which may be thrown from the wheel, and to protect the tires from accidental spillage of bitumen. Each fender shall withstand a 200-lb load over any area 10 inches square. Each fender shall have a full backplate to protect the tires from the heat radiated from the kettle if other methods of heat shielding for tire protection are not employed in the kettle design.

3.10.5 Wheels and rims. The trailer wheels shall be the manufacturer's standard. Rim and tire ratings shall conform to the TRA recommendations for the type and size of tires furnished.

3.10.5.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire measured at each wheel, at the ground, with the kettle loaded to capacity. Tires shall be of not less than 100 level quality. Manufacturer's current size and type tire conforming to specified requirements shall be furnished.

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3.10.5.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of proper size for tires furnished. When mounted on other than drop center rims, tire flaps shall be installed.

3.11 Brakes. When specified (see 6.2), kettles shall be furnished with electric or hydraulic brakes as specified herein.

3.11.1 Electric brakes. When electric brakes are required, the kettle shall be furnished with a 12-volt (V) electric brake system wired in accordance with 3.11.1.1. The system shall include a controller kit for mounting on the dash of the towing vehicle, a controller kit for synchronizing the trailer electric brakes with the brakes of the towing vehicle, and an emergency breakaway kit. Kits shall be complete with all necessary parts (except stowage battery for breakaway kit) and installation instructions.

3.11.1.1 Brake wiring. Electric brakes shall be wired to the auxiliary circuit of the seven-conductor electrical connector conforming to SAE J560. All wiring shall be properly mounted and secured with clamps. Jumper cable shall be heavy wall thermoplastic insulated cable conforming to SAE J1067.

3.11.2 Hydraulic brakes. When hydraulic brakes are required, the kettle shall be equipped with a hydraulic surge type braking system. The surge brake system shall incorporate the ability to allow the kettle to be backed up by the towing vehicle without the brakes being applied. The brake actuator shall be of the hitch surge type, tongue mounted, and shall be Atwood, Bendix, Dico Brand, or equal. The kettle shall be equipped with brakes of not less than one axle when tandem axle kettles are provided.

3.12 Lighting. When specified (see 6.2), an electrical lighting system shall be provided. The electrical lighting system shall be 12V. All vehicle lights, reflectors, and wiring (including electric brakes) shall conform to DoT Federal Motor Carrier Safety Regulation 393.15, 393.19, 393.20, 393.22, 393.23, 393.25, 393.26 (a, b, c, and d), and 393.27 through 393.33. Turn signals conforming to SAE J588 shall be furnished.

3.13 Toolbox. A toolbox shall be provided. The toolbox shall be large enough to store all tools required for field service or maintenance. The toolbox shall be made of 0.0747 inch (U.S. revised standard gage No. 14) thick steel with a hinged lid and trunk drawbolt. The toolbox shall be mounted in a protected, accessible location.

3.14 Lifting and tiedown attachments. When specified (see 6.2), the kettle shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the kettle. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the kettle on the carrier when shipped.

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3.15 Radio interference suppression. When specified (see 6.2), the kettle shall be suppressed for radio interference in accordance with SAE J551.

3.16 Identification plate. An identification plate will be furnished by the contracting officer for each kettle. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each kettle in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.17 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel or polyurethane. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.18 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.19 Workmanship.

3.19.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.19.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.19.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter

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of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.19.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.19.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings' ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on a kettle when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4.1 through 4.4.7 and when specified, the preproduction pack inspection of 4.5 (see 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.8, and the packing inspection of 4.5.

4.3 Examination. Each kettle shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified

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requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Road. With the melting compartment filled to a safe towing level with liquid asphalt, or suitable substitute, the engine fuel tank filled to capacity, and with full-burner fuel tanks(s) mounted on the trailer chassis, tow the kettle for not less than 20 miles over paved roads at legal speeds up to 55 mph. Leaks, cracked joints or weldments, or permanent deformation to any part of the kettle shall constitute failure of this test.

4.4.2 Capacity. Determine the capacity, including expansion allowance (see 3.9.1 and 3.9.2), of the melting tank. A melting tank capacity less than that specified for the applicable size, shall constitute failure of this test.

4.4.3 Grille. Kettles furnished with grille (see 3.9.6), shall be subjected to the following grille test. Place a 55-gallon drum filled with asphalt cement or a comparable weight on the grille. Bends, breaks, or permanent deformation of the grille or any part of the kettle shall constitute failure of this test.

4.4.4 Burner(s). Ignite and operate the burner(s) for one hour. Malfunction of the burner(s), burner controls, or smoking in excess of 20 percent opacity (see 3.7), after unit has reached normal operating conditions shall constitute failure of this test.

4.4.5 Performance. Fill the melting tank to operating capacity with asphalt as specified in 3.9.1. Allow the asphalt to obtain ambient temperature before testing. Ignite the burner(s) and heat the asphalt to 325°F for type I kettles, 525°F for type II kettles. Controls shall maintain temperature for not less than five hours with a variance of not over ±25°F. Addition of cold material is acceptable during testing. The liquid asphalt shall be circulated during warm-up. The kettle shall be operated to demonstrate the ability to perform the requirements listed below. Inability to accomplish any requirement to the extent specified below shall constitute failure of this test:

a. Pump cleaning solvent through the hand spray attachment without the solvent entering the melting tank. This test applies to non-submerged pump systems.
b. Pump liquid asphalt through the spray nozzle at a rate of not less than 10 gpm.
c. Produce not less than 30 psi at the spray nozzle.
d. Circulate liquid asphalt being heated in the melting tank.
e. Pump liquid asphalt through 1-1/2 inch thinwall tubing to a height of not less than 75 feet. This test applies to type II kettles only.

4.4.6 Lifting and tying down attachment tests. When required, the lifting and tying down attachments shall be tested to determine conformance with 3.14.

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4.4.7 Measurement of electromagnetic radiation. To determine conformance to 3.15, electromagnetic radiation shall be measured in accordance with SAE J551. When suppressed to conform to 3.15, the manufacturer may, upon approval of the contracting officer, furnish a certification in lieu of the test that kettles meet the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.4.8 Functional test. Each kettle shall be serviced in accordance with the manufacturer's recommendation prior to testing. Test each kettle melting compartment for leaks. Ignite the burner(s) and operate for a period of time necessary to determine that the burner(s) and all burner controls function properly. Determine that the engine and all its controls operate properly. The kettle pumping system shall be tested with a suitable medium to simulate actual operating conditions to assure proper performance. Leaks, malfunction of burner(s) and burner controls, or malfunction of the pump, engine, and engine controls, which cannot be corrected by minor adjustment shall constitute failure of this test.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of Section 4 of MIL-D-771. The inspection shall consist of the quality conformance inspection; and when specified (see 6.2), a preproduction pack shall be furnished for examination and tested within the time frame required.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-D-771 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. Type I kettles are intended for use in melting and dispensing all types of bituminous material for maintenance of roadways, runways, and parking aprons. Type II kettles are intended for use in melting and pumping to the roof all types of roofing cements.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, style, and size of kettle required (see 1.2).
c. Specifications, standards, and handbooks (see 2.1.1).
d. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
e. When a grille shall be furnished for type I kettles of 165-gallon capacity and larger (see 3.9.6).
f. Number of lengths, diameter, number of elbows, connectors, and couplings required (see 3.9.14).
g. When a tongue mounted screw jack shall be provided (see 3.10.2).
h. When electric or hydraulic brakes are required (see 3.11).
i. When an electrical lighting system is required (see 3.12).
j. When lifting and tiedown attachments as required (see 3.14).
k. When radio interference is to be suppressed (see 3.15).

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l. Color of finish paint coat required (see 3.17).
m. When lubrication shall be other than the manufacturer's standard practice (see 3.18).
n. When a preproduction pack inspection is required and time frame required for submission (see 4.2.1 and 4.5).
o. Level of preservation and level of packing required (see 5.1).

* 6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

* 6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

* 6.5 Subject term (key word) listing.

Bitumen heating
Kerosene fueled
Liquified petroleum gas
Paving
Roofing
Tube fired

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-K-28532C(YD)	911118

3. DOCUMENT TITLE
KETTLES, HEATING, BITUMEN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDING OFFICER	(805) 982-5464	551-5464
c. ADDRESS (Include Zip Code) NAVAL CONSTRUCTION BATTALION CENTER CESO (CODE 1564L) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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